

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: CHEMISTRY

2. Subject Name: Chemistry of Transition Metals L-T-P: 4-0-0 Credits:4

3. Pre-requisites: NONE

4. Syllabus and reference books:

Syllabus:

1. General properties of transition elements, Term symbols. (6 L)
2. Coordination Chemistry: Constitution and geometry; structure and symmetry, representative ligands and nomenclature, chelate and macrocyclic effect, steric effects, isomerism and chirality, factors affecting ligand field stabilization energies, bonding and electronic structure (CFT treatment only), color and magnetic properties. (24 L)
3. Multiple bonded metal complexes of Re, Mo, and others: Bonding and structure. (2 L)
4. Essential features of f-block elements: the valence orbitals, lanthanide contraction, ion exchange separation, electronic configuration, colour and magnetism; physical properties and applications. (6 L)
5. Fundamental of organometallics: Introduction to organometallic chemistry, low valent and π -acid ligands, 18 electron rule and its applications, bonding, and structure. (6 L)
6. Fundamental of bioinorganic chemistry: Ligands and metal ions in living systems, iron and copper-based oxygen transport systems and its model compounds. (6 L)

Reference Books:

- (1) Inorganic Chemistry by Shriver, Weller, Overton, Rourke, Armstrong; 6th Edn. W H Freeman and Co.
- (2) Inorganic Chemistry-Principles of Structure and Reactivity. J. E. Huheey, E. A. Keiter, R. L. Keiter, 4th Edn. Harper-Collins, NY, 1993.
- (3) Advanced Inorganic Chemistry, F. Albert Cotton, Geoffrey Wilkinson, Carlos A. Murillo, Manfred Bochmann, Wiley 1997
- (4) Inorganic Chemistry, Catherine E. Housecroft, Pearson; 5th edition 2018
- (5) Lanthanide and Actinide Chemistry, Simon Cotton, Wiley, 2016

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	General properties of transition elements	06
2.	Coordination Chemistry	24
3.	Multiple bonded metal complexes of Re, Mo, and others	02
4.	Essential features of f-block elements	06
5.	Fundamental of organometallics	06
6.	Fundamental of bioinorganic chemistry	06
Total number of hours		50